CS 532, Spring 2019: Quiz 5

NAME:

Problem 1. Consider a pair of rectified cameras with baseline b=10 cm, focal length equal to f= 5 pixels. If the true disparity of a given pixel is d^*=3, answer the following:

- What is the depth estimation error if we underestimate d^* by one pixel (i.e. $\hat{d} = d^* - 1$)

- What is the depth estimation error if we overestimate d^* by one pixel (i.e. $\hat{d} = d^* + 1$)

- Would these depth estimation error values remain constant for larger disparity values? Justify your answer